

EXAM 7 – SPRING 2013

3. (5 points)

Given the following information:

	<u>Cumulative Paid Loss (\$000)</u>		
<u>Accident Year</u>	<u>12</u>	<u>24</u>	<u>36</u>
2010	2,750	4,250	5,100
2011	2,700	4,300	
2012	2,900		

The expected accident year loss emergence pattern (growth function) is approximated by a Weibull function of the form:

$$G(x|\omega, \theta) = 1 - \exp(-(x/\theta)^\omega)$$

Parameter estimates are: $\omega = 1.5$ and $\theta = 20$

a. (3 points)

Calculate the process standard deviation of the reserve estimate for accident years 2010 through 2012 using the LDF method.

b. (2 points)

Graph the normalized residuals plotted against the increment age of loss emergence. Based on your graphical results, discuss the appropriateness of the Weibull model above.

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Question 3 Sample Answer

Solution 1

a)

$$\text{Process st dev} = \sqrt{R\sigma^2} \quad \sigma^2 = 1/(n-p) \sum (c - \mu)^2 / \mu \quad n=6 \quad p=2+3=5$$

$$G(6) = 1 - e^{-(6/20)^{1.5}} = .1515$$

$\uparrow$ $\uparrow$
 G,W Ult 2010, Ult 2011,
 Ult 2012

$$G(18) = .5742$$

$$G(30) = .8407$$

* Check if truncation is needed:

- Extend Δ out 3 years $G(66) = .9975$
- $G(66)$ is reasonably close to 1 and so the function does not need to be truncated

<u>AY</u>	<u>G(x)</u>	<u>LDF = 1/G(x)</u>	<u>Ult = Cum Pd × LDF</u>	<u>Reserve</u>
2010	0.8407	1.1895	6,066.4	966.4
2011	0.5742	1.7416	7,488.7	3,188.7
2012	0.1515	6.6007	19,142	<u>16,242</u>
				20,397.1

	avg age:	(6)	(18)	(30)
<u>expected</u>		12	24	36
10		919	2564.3	1616.7
11		1134.5	3165.5	1995.7
12		2900	8091.3	5101.3

$$\mu = [G(y) - G(x)] \text{ Ult AY}$$

$$1995.7 = (.8407 - .5742)(7488.7)$$

actual

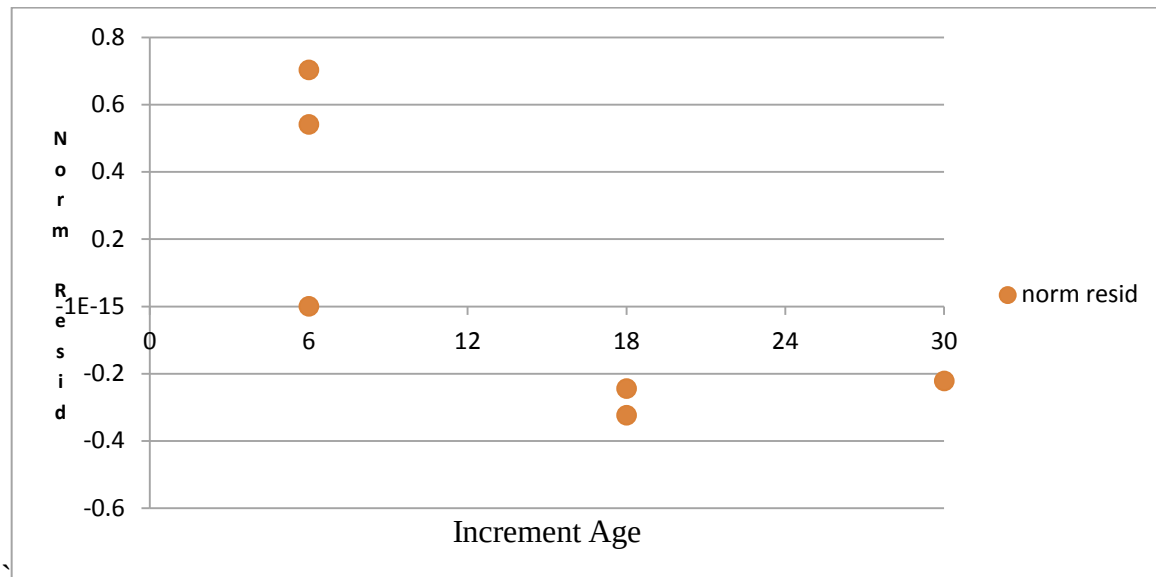
10	2750	1500	850
11	2700	1600	
12	2900		↖ 4300-2700

$$\sigma^2 = 1/(6-5) \times \left[\frac{(2750-919)^2}{919} + \frac{(2700-1134.5)^2}{1134.5} + \dots \right] = 7387.84$$

$$\text{process standard dev} = \sqrt{R\sigma^2} = \sqrt{7387.84 * 20397.1} = 12,275.607$$

$$\text{b) } \varepsilon = (c - \mu) / \sqrt{\mu \sigma^2}$$

6	18	30
0.7027	-0.245	-0.222
0.5407	-0.324	↑
		(850-
0		1616.7)/(7387.4×1616.7)^0.5



*expect points randomly scattered around zero, but it appears that the residuals at age 6 are all positive (μ is underestimating) then the residuals are negative at later ages (μ is over estimating). Thus Weibull model used may not be appropriate for this loss experience.

Solution 2

a) Avg Age

<u>Avg Age</u>	<u>G(Age)</u>
6	$1 - e^{-(6/20)^{1.5}} = .152$
18	$= .574$
30	$= .841$
$12(6) - 6 = 66$	$= .998$ Truncate at twice the triangle or $3(2) = 6$ years

AY	Paid to Date	% paid	LDF- Ultimate	LDF Truncated
12	2,900	0.152	$1/.152 = 6.58$	$6.58(.998) = 6.57$
11	4,300	0.574	1.74	1.736
10	5,100	0.841	1.19	1.188

<u>AY</u>	<u>Ult Loss</u>	<u>Reserve</u>
12	6.58×2900-19053	19053-2900=16153
11	1.736×4300=7465	3165
10	1.188×5100=6059	959
		Total=20277 <-- Total Reserves

Incremental Expected Triangle = Ult (%)
paid at age)

<u>AY</u>	<u>12</u>	<u>24</u>	<u>36</u>
10	921	2557	1618
11	1135	3150	
12	2896		

$$6059 \times (.152) = 921$$

Actual Incremental
Triangle

<u>AY</u>	<u>12</u>	<u>24</u>	<u>36</u>
10	2750	1500	850
11	2700	1600	
12	2900		

(actual - expected)² / expected

<u>AY</u>	<u>12</u>	<u>24</u>	<u>36</u>
10	3632	437	365
11	2158	763	
12	0.006		

Sum of triangle = 7358

Total Reserves = 20,277

Process Variance = 20277(σ^2)

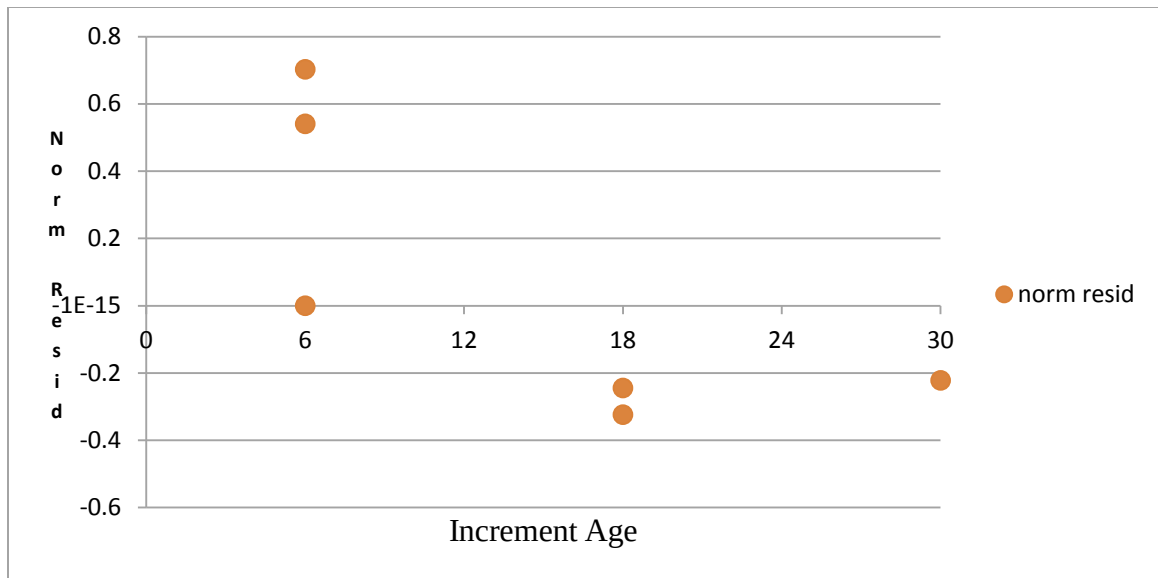
$$\sigma^2 = 1/(n-p) \sum (A - E)^2 / E = 1/(6-5)(7335) = 7335$$

$$\text{Process standard deviation} = \sqrt{20277(7335)} = 12196$$

$$\text{b) normalized residual} = (\text{actual-expected}) / (\sigma(\text{expected})^{0.5}) \quad \sigma = \sqrt{7335} = 85.6$$

Residual
Triangle

<u>AY</u>	<u>12</u>	<u>24</u>	<u>36</u>
10	0.704	-0.244	-0.223
11	0.543	-0.323	
12	0		



If the Weibull curve were appropriate, we would see residual randomly scattered around 0 at each age. In our graph, there is a clear trend from positive to negative as age increases. This means this Weibull model is not appropriate.

Examiner Comment

- a) The candidate was asked to determine the process standard deviation given a specific model. To get there, several basic calculations were needed such as determining ultimate losses given specific data and a specific curve, resulting reserves, and the chi squared formula – including corresponding degrees of freedom. Common errors included using cumulative instead of incremental actual and fitted data triangles, not recognizing the use of the average age in the application of the Weibull model, not correctly determining the degrees of freedom to use, and general math errors. Candidates often did not calculate the process standard deviation correctly - either incorrectly taking the square root of σ^2 or calculating the correct process variance but forgetting that the standard deviation was requested.

Truncation was not specifically mentioned in the question and could be subject to candidate interpretation of model results. Both truncated and untruncated approaches were accepted for full credit.

- b) The candidate is asked to calculate and graph normalized residual using the results from part a). A very common error in the determination of the normalized residuals was to fail to divide by the σ resulting in a set of residuals incorrectly scaled. Another common mistake was to take the square root of the chi-squared results from part a) which failed to

recognize the possibility of negative residuals. Candidates generally graphed their calculated residuals correctly, but a common mistake observed was to graph the residual against incremental loss size which was not what the question specified. In the evaluation of the graph, most candidates recognized that the Weibull was not appropriate given the data and resulting residuals; however, explanations as to why were oftentimes incomplete or non-existent.